THE MIRROR LAW (mLA): WHEN THE NORMALIZED MAGMOID IS THE QUOTIENT $\mathcal{A}/\ker K$, AND THE TWO CLASSICAL ROADS TO ASSOCIATIVITY

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ABSTRACT. In the positive home of the normalization defect, a *normalizer* K on an associative category $\mathcal{A}$ satisfies the left-absorption law (LA) $K(K(x)\cdot c) = K(x\cdot c)$ and generates a magmoid $\mathbf{Norm}_K = \operatorname{Im} K$ with product $f \# g := K(f\cdot g)$; this magmoid is associative exactly when the two-sided law ($\mathsf{RA}_\#$) holds, and *handedness* is (LA)$\wedge\neg$($\mathsf{RA}_\#$). Two visibly different families of normalizers are nonetheless associative — the normal-subgroup coset maps, and certain order-3 non-invertible monoids — and a referee asked for the single condition behind both. We answer it. Define the *mirror law* (mLA) $K(c\cdot K(x)) = K(c\cdot x)$, the two arguments of (LA) swapped. We prove three things. (1) For a normalizer, (mLA) alone forces ($\mathsf{RA}_\#$), and (LA)$\wedge$(mLA) makes the kernel relation $a \sim_K b :\Leftrightarrow K(a) = K(b)$ a *two-sided congruence*, so that $\mathbf{Norm}_K \cong \mathcal{A}/\!\sim_K$ as categories — the magmoid is literally $\mathcal{A}$ with $\sim_K$-equal arrows identified; indeed (mLA) is *equivalent* to "$\sim_K$ is a left congruence". (2) (mLA) is strictly stronger than ($\mathsf{RA}_\#$), and incomparable with closure of $\operatorname{Im} K$: each holds while the other fails, so they are two genuinely different sufficient conditions, forming a diamond over associativity. (3) For a group G with K the right-coset-representative map of $H \le G$, (mLA)$\Leftrightarrow H \triangleleft G$ (then $\mathbf{Norm}_K \cong G/H$); without normality, writing $g\cdot h = \varphi(g,h)\cdot(g\#h)$ with $\varphi \in H$, the transversal product is associative $\Leftrightarrow f\varphi(g,h)f^{-1} \in H$ for all f, g, h in the transversal. Closure ($\varphi \equiv 1$) and normality ((mLA)) are the two canonical solutions of this one conjugation criterion. The results are proved by hand, corroborated by exhaustive enumeration over all monoids of order ≤ 3 and all (H, T) pairs of S_3 and S_4 (36 000+ pairs, zero mismatches), and the finite core is machine-checked axiom-free in Lean 4.

1. The question, and the setting

This note is a companion to the triangulation unit (*What normalization's non-associativity is, and is not*) and its closure criterion. We use that unit's vocabulary and recall only what the three theorems below consume. Composition is diagrammatic: $f \cdot g$ reads "f then g".

Definition 1.1 (normalizer, normalized magmoid)**.** A *normalizer* on an associative category $(\mathcal{A}, \cdot, \mathrm{id})$ is a family of endomaps $K_{X,Y} \colon \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ satisfying, for all composable x, c,

(K1) $K(Ka) = Ka$ (idempotent),
(U) $K(\mathrm{id}_X) = \mathrm{id}_X$ (unital),
(LA) $K\big(K(x)\cdot c\big) = K(x\cdot c)$ (left absorption).

Its *normalized magmoid* $\mathbf{Norm}_K$ has the same objects, hom-sets $\operatorname{Im} K = \{a : K(a) = a\}$ (equal by (K1)), product $f \# g := K(f\cdot g)$, and units id_X. We call $\operatorname{Im} K$ *closed* when $f, g \in \operatorname{Im} K \Rightarrow f\cdot g \in \operatorname{Im} K$, i.e. $K(f\cdot g) = f\cdot g$.

The magmoid $\mathbf{Norm}_K$ is a category-minus-associativity; the triangulation unit records (its Theorem B) exactly when associativity returns.

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[0]MacBeth (scotmacbeth) · 2026-10-23 · For: Rick · content commit `70b090c`. A self-contained write-up of the registry node `mirror-LA-sufficient-for-associativity` (proof `proofs/2026-10-22-mirror-LA-sufficient-for-associativity.md`), under the peer-reviewed positive-home parent. Answers SHOULD S1(b) of the uid 239 referee report on the closure criterion. The finite core is Lean-certified (`lean-mirror-LA-oracle`). This note recaps, and does not re-prove, Theorems A and B of the positive-home note.

Proposition 1.2 (Theorem B of the positive-home note, recalled)**.** $\mathbf{Norm}_K$ *is associative if and only if the two-sided law*

$$(\mathsf{RA}_{\#}): \qquad K\big(f \cdot K(g \cdot h)\big) = K(f \cdot g \cdot h) \qquad (f, g, h \in \operatorname{Im} K)$$

holds. The normalizer K *is* handed *when* (LA) *holds but* ($\mathsf{RA}_{\#}$) *fails; handedness is the non-associativity of* $\mathbf{Norm}_K$*.*

Two families of normalizers turn out to be associative for reasons that look nothing alike. The first: $\mathcal{A} = G$ a group, K the right-coset-representative map of a *normal* subgroup, where $\mathbf{Norm}_K \cong G/H$. The second: small non-invertible monoids whose normalizer has *closed* image (the closure criterion's positive side). The referee's question (uid 239, SHOULD S1(b)) was whether one condition sits behind both. It does.

Definition 1.3 (kernel relation, mirror law)**.** On each hom-set, $a \sim_K b :\Leftrightarrow K(a) = K(b)$; write $[a]$ for the class and $\mathcal{A}/\!\sim_K$ for the quotient graph with hom-sets $\mathcal{A}(X,Y)/\!\sim_K$. The normalizer K satisfies the *mirror law* — (LA) with its two arguments swapped — if

$$(\mathsf{mLA}): \qquad K\big(c \cdot K(x)\big) = K(c \cdot x) \qquad \text{for all composable } c, x \in \mathcal{A}.$$

Remark 1.4 ((mLA) is a genuine constraint). As with (LA), the variable x ranges over *all* of $\mathcal{A}$, in particular over un-normalized products $x = g \cdot h$ of $\mathbf{Norm}_K$-arrows. Where $K(x) \neq x$, (mLA) asserts that the inner normalization is invisible to a *left* multiply-then-normalize. It is *not* the triviality $K(c \cdot K(x)) = K(c \cdot x)$ for $x \in \operatorname{Im} K$ (where both sides are $K(c \cdot x)$ by (K1)). This is the mirror of Remark 1.3 of the positive-home note.

The three theorems say: (§2) (mLA) is exactly the missing left-handed half, and it upgrades associativity to a quotient; (§3) it is strictly stronger than associativity and incomparable with closure; (§4) over groups it *is* normality, and it names the exact boundary.

2. The mirror law makes $\mathbf{Norm}_K$ the quotient $\mathcal{A}/\ker K$

Theorem 2.1. *Let* K *be a normalizer satisfying* (mLA)*. Then:*

1. ($\mathsf{RA}_{\#}$) *holds; hence* $\mathbf{Norm}_K$ *is associative.*
2. $\sim_K$ *is a* two-sided congruence *on* $\mathcal{A}$*: if* $a \sim_K a'$ *and* $b \sim_K b'$ *(composable) then* $a \cdot b \sim_K a' \cdot b'$*. Consequently* $\mathcal{A}/\!\sim_K$ *is a category under* $[a] \cdot [b] := [a \cdot b]$*, units* $[\mathrm{id}_X]$*.*
3. $\Phi\colon \mathbf{Norm}_K \to \mathcal{A}/\!\sim_K$*,* $\Phi(f) = [f]$*, is an* isomorphism of categories*, with inverse* $\Psi([a]) = K(a)$*. In particular* $\mathbf{Norm}_K \cong \mathcal{A}/\!\sim_K$*.*

The key idea is that a normalizer always carries *one* half of a congruence — (LA) is the right-handed half — and (mLA) supplies the other. With both halves the kernel is a congruence, and the quotient of a category is a category, automatically associative; the normalize-then-compose product # is *forced* to agree with the quotient product because $K(f \cdot g) \sim_K f \cdot g$. That is the whole mechanism.

Proof. **(1).** Instantiate (mLA) at $c := f$ and $x := g \cdot h$, with $f, g, h \in \operatorname{Im} K$:

$$K\big(f \cdot K(g \cdot h)\big) = K\big(f \cdot (g \cdot h)\big) = K(f \cdot g \cdot h).$$

The left-hand side is the left-hand side of ($\mathsf{RA}_{\#}$); the right-hand side is its right-hand side. So ($\mathsf{RA}_{\#}$) holds, and $\mathbf{Norm}_K$ is associative by Proposition 1.2. (This step uses (mLA) only; (LA) enters in part (2).)

(2). We establish the two one-sided compatibilities separately.

Right-compatibility, from (LA)*.* Read (LA) backwards as $K(x \cdot c) = K(K(x) \cdot c)$. If $K(a) = K(a')$ with $a, a' \in \mathcal{A}(X,Y)$ and $b \in \mathcal{A}(Y,Z)$, then

$$K(a \cdot b) = K\big(K(a) \cdot b\big) = K\big(K(a') \cdot b\big) = K(a' \cdot b),$$

so $a \cdot b \sim_K a' \cdot b$. This holds for *every* normalizer.

Left-compatibility, from (mLA). If $K(b) = K(b')$ with $b, b' \in \mathcal{A}(Y, Z)$ and $a \in \mathcal{A}(X, Y)$, then applying (mLA) with $c := a$,

$$K(a \cdot b) = K\big(a \cdot K(b)\big) = K\big(a \cdot K(b')\big) = K(a \cdot b'),$$

so $a \cdot b \sim_K a \cdot b'$.

Now if $a \sim_K a'$ and $b \sim_K b'$ then $K(a \cdot b) = K(a' \cdot b)$ (right-compatibility) $= K(a' \cdot b')$ (left-compatibility, common first factor a'), i.e. $a \cdot b \sim_K a' \cdot b'$. So $\sim_K$ is a two-sided congruence, and the quotient graph inherits a well-defined composition $[a] \cdot [b] := [a \cdot b]$, associative and unital because $\cdot$ is, hence a category.

(3). Ψ is well-defined: $a \sim_K a' \Rightarrow K(a) = K(a') \in \operatorname{Im} K = \mathbf{Norm}_K$. The two maps are mutually inverse:

$$\Psi(\Phi(f)) = \Psi([f]) = K(f) = f \quad (f \in \operatorname{Im} K), \qquad \Phi(\Psi([a])) = \Phi(K(a)) = [K(a)] = [a],$$

the last because $K(K(a)) = K(a)$ gives $K(a) \sim_K a$ by (K1). And Φ is functorial:

$$\Phi(f \# g) = \big[K(f \cdot g)\big] = [f \cdot g] = [f] \cdot [g] = \Phi(f) \cdot \Phi(g), \qquad \Phi(\mathrm{id}_X) = [\mathrm{id}_X],$$

using $K(f \cdot g) \sim_K f \cdot g$ for the second equality. Being bijective on objects and on each hom-set, Φ is an isomorphism of categories. □

Remark 2.2 ((mLA) is exactly "$\sim_K$ a left congruence"). Theorem 2.1(2) shows (mLA) $\Rightarrow \sim_K$ a left congruence. The converse holds using only (K1): if $\sim_K$ is a left congruence, then since $K(x) \sim_K x$, left-compatibility with first factor c gives $K(c \cdot K(x)) = K(c \cdot x)$, which is (mLA). Hence

$$\text{(mLA)} \iff \sim_K \text{ is a left congruence,}$$

independently of (LA); dually, (LA) $\Leftrightarrow \sim_K$ is a *right* congruence. This is the reading Theorem 4.1 uses.

Remark 2.3 (what Theorem 2.1 adds to Theorem B). Proposition 1.2 says *when* $\mathbf{Norm}_K$ is associative; Theorem 2.1 says *what it then is.* (RA$_\#$) is the bare ternary associativity equation; (mLA) is the stronger, more legible statement that the kernel is a congruence, so $\mathbf{Norm}_K$ is $\mathcal{A}$ with $\sim_K$-equal arrows identified. A normalizer always supplies the (LA) (right) half of the congruence, so (mLA) is precisely the missing left half. Both associative families of §1 will turn out to be quotients $\mathcal{A}/\!\sim_K$.

3. Strictly stronger than associativity, and incomparable with closure

(mLA) is a *sufficient* condition for associativity. So is closure of $\operatorname{Im} K$ (the triangulation unit's closure criterion). Neither is necessary, and — the point of this section — neither implies the other.

Theorem 3.1. *There is a normalizer with (LA)∧(RA$_\#$)∧¬(mLA); so (mLA) ⇒ (RA$_\#$) is a proper implication. Moreover (mLA) and closure of* $\operatorname{Im} K$ *are* incomparable*: each can hold while the other fails. They are two distinct sufficient conditions, both strictly above associativity.*

Proof. A separator with (RA$_\#$)∧¬(mLA) and $\operatorname{Im} K$ *closed.* Let $M = \{0, 1, 2\}$ with 0 the unit and multiplication (rows = left factor)

$\cdot$	0	1	2
0	0	1	2
1	1	1	2
2	2	1	2

M is associative and $1, 2$ are non-invertible (both checked exactly). Take $K(0) = 0$, $K(1) = 0$, $K(2) = 2$, so $\operatorname{Im} K = \{0, 2\}$. Then (K1), (U), (LA) hold (checked over all 3×3 pairs), and $\operatorname{Im} K$ is *closed* ($0 \cdot 0 = 0$, $0 \cdot 2 = 2$, $2 \cdot 0 = 2$, $2 \cdot 2 = 2$), so (RA$_\#$) holds and $\mathbf{Norm}_K$ — the two-element monoid $\{0, 2\}$ with 2 absorbing — is associative. But (mLA) fails at $c = 2$, $x = 1$:

$$K\big(2 \cdot K(1)\big) = K(2 \cdot 0) = K(2) = 2, \qquad K(2 \cdot 1) = K(1) = 0, \qquad 2 \neq 0.$$

Thus $(\mathsf{RA}_{\#}) \wedge \neg(\mathsf{mLA})$ — so $(\mathsf{mLA}) \Rightarrow (\mathsf{RA}_{\#})$ is proper — and, since $\operatorname{Im} K$ is closed here, *closure does not imply (mLA).*

A separator with (mLA) but $\operatorname{Im} K$ *not closed.* The order-3 converse witness of the closure note ($M = \{e, a, b\}$, $a \cdot a = b$, $a \cdot b = b \cdot a = b \cdot b = a$, with $K(e) = e$, $K(a) = a$, $K(b) = a$) has $\sim_K$-classes $\{e\}, \{a, b\}$, a two-sided congruence, so (mLA) holds; yet $\operatorname{Im} K = \{e, a\}$ is not closed because $a \cdot a = b \notin \operatorname{Im} K$. So *(mLA) does not imply closure.* The two conditions are incomparable. □

The failure in the first separator is diagnostic. Closure says the *image* is a subcategory; (mLA) says the *kernel* $\sim_K$ is a left congruence. Here the classes are $\{0, 1\}$ and $\{2\}$, and left-multiplying the class $\{0, 1\}$ by 2 sends $2 \cdot 0 = 2$ into class $\{2\}$ but $2 \cdot 1 = 1$ into class $\{0, 1\}$: left multiplication by 2 does not respect $\sim_K$. A closed image is blind to this; (mLA) is not.

Remark 3.2 (the diamond). Theorem 3.1 draws a diamond over associativity:

$$\mathbf{Norm}_K \text{ associative } = (\mathsf{RA}_{\#})$$

$$\nearrow \qquad\qquad \nwarrow$$

$(\mathsf{mLA}) : \sim_K$ a left congruence [(mLA) holds, not closed] — closure: $\operatorname{Im} K$ a subcategory [closed, (mLA) fails]

Each upward arrow is strict and the two upper conditions are logically independent. Over groups (§4) they are the two classical mechanisms: $(\mathsf{mLA}) \Leftrightarrow H$ normal, closure $\Leftrightarrow T$ a subgroup. Handedness $= (\mathsf{LA}) \wedge \neg(\mathsf{RA}_{\#})$ therefore requires both $\neg(\mathsf{mLA})$ and ¬closure — each is necessary for associativity to fail, neither alone is the obstruction.

4. Over a group: the mirror law is normality

Let G be a group, $H \leq G$ a subgroup, and T a *right transversal* (one representative per right coset Hg, with $\mathrm{id} \in T$). Let $K\colon G \to G$ send g to the unique $t \in T$ with $Hg = Ht$; then $\operatorname{Im} K = T$ and $\#$ is the transversal (right-loop) product $t \# t' = K(t \cdot t')$.

Two facts are immediate. First, (LA) holds for *every* transversal rep map: since $K(x) \in Hx$, we have $H \cdot K(x) = H \cdot x$, hence $H \cdot (K(x) \cdot c) = H \cdot (x \cdot c)$ and $K(K(x) \cdot c) = K(x \cdot c)$. Second, $K(a) = K(b) \Leftrightarrow Ha = Hb$, so the classes of $\sim_K$ are exactly the *right cosets* of H.

Theorem 4.1. *With K the right-coset-representative map of $H \leq G$: (mLA) holds $\Leftrightarrow H \triangleleft G$. When $H \triangleleft G$, Theorem 2.1(3) specializes to the classical isomorphism $(T, \#) \cong G/H$.*

Proof. By Remark 2.2, $(\mathsf{mLA}) \Leftrightarrow \sim_K$ is a left congruence. The $\sim_K$-classes are right cosets, so left-compatibility reads: $Ha = Hb \Rightarrow H(ca) = H(cb)$ for all c. Equivalently, for all a, b, c: $ab^{-1} \in H \Rightarrow (ca)(cb)^{-1} = c(ab^{-1})c^{-1} \in H$. Letting ab^{-1} range over all of H and c over G, this is exactly $cHc^{-1} \subseteq H$ for all c, i.e. $H \triangleleft G$. When H is normal, $\sim_K$ is equality of (two-sided) cosets, so $\mathcal{A}/{\sim_K} = G/H$ and $\Phi\colon t \mapsto Ht$ is the group isomorphism $(T, \#) \cong G/H$. □

So the normal-subgroup family of associative transversals *is* the (mLA) family. The remaining associative transversals — associative with H not normal — are governed by an explicit ternary criterion.

Theorem 4.2 (the exact boundary)**.** *For $f, g, h \in T$ write the de-normalization $g \cdot h = \varphi(g, h) \cdot (g \# h)$ with $\varphi(g, h) := (g \cdot h)(g \# h)^{-1} \in H$. Then $(T, \#)$ is associative $\Leftrightarrow f\,\varphi(g, h)\,f^{-1} \in H$ for all $f, g, h \in T$.*

Proof. Since $g \# h = K(g \cdot h) \in Hg \cdot h$, indeed $\varphi(g, h) \in H$ and $g \# h = \varphi(g, h)^{-1} \cdot (g \cdot h)$. By (LA) and Theorem A of the positive-home note, $(f \# g) \# h = K(f \cdot g \cdot h)$; and

$$f \# (g \# h) = K\big(f \cdot (g \# h)\big) = K\big(f \cdot \varphi(g, h)^{-1} \cdot g \cdot h\big).$$

Associativity is $K(f \cdot g \cdot h) = K(f \cdot \varphi(g,h)^{-1} \cdot g \cdot h)$ for all $f, g, h \in T$. Using $K(x) = K(y) \Leftrightarrow xy^{-1} \in H$, this is

$$(f \cdot g \cdot h)\big(f \cdot \varphi(g,h)^{-1} \cdot g \cdot h\big)^{-1} = f \cdot g \cdot h \cdot (g \cdot h)^{-1} \cdot \varphi(g,h) \cdot f^{-1} = f\,\varphi(g,h)\,f^{-1} \in H$$

for all $f, g, h \in T$. □

Remark 4.3 (the criterion is the diamond). The criterion $f\varphi(g,h)f^{-1} \in H$ is satisfied in two canonical ways, matching Remark 3.2:

- *closure / T a subgroup:* then $g \cdot h \in T$, so $g \# h = g \cdot h$ and $\varphi(g,h) = \mathrm{id}$; the criterion is vacuous.
- *normality / (mLA):* $\varphi(g,h) \in H$ and $H \triangleleft G$, so $f\varphi(g,h)f^{-1} \in H$ automatically.

The criterion is strictly weaker than either: it permits a nontrivial φ, valued in a non-normal H, that merely stays in H under conjugation by the finitely many $f \in T$. That strictly-between regime is the honest content of "associative without normality or closure". This mirrors the split-vs-cocycle dichotomy of the weld H^2 program: closure is the trivial-cocycle locus ($\varphi \equiv \mathrm{id}$), normality a cocycle-stability locus, and associativity strictly below both.

5. Verification and scope

Lean (finite core, axiom-free). The leaf `lean-mirror-LA-oracle` (`lean/2026-10-22-mirror-LA-oracle.lean`, Lean 4, Mathlib-free) machine-checks the decidable content: on the order-4 handed model, (mLA) fails (as it must, being handed); (mLA) forces ($\mathsf{RA}_\#$) and #-associativity over *all* monoids of order ≤ 3 and all idempotent unit-fixing K (zero counterexamples); (mLA) $\Leftrightarrow\!\sim_K$ a left congruence over the same range (zero mismatches, Remark 2.2); the strictness separator exists at order 3 and not below; and the explicit separator of §3 (closed image, ($\mathsf{RA}_\#$) true, (mLA) false at $c = 2, x = 1$) is certified. The bundled theorem reports *no* axiom dependence — not even `propext`; every `decide` reduces to `rfl`. We cite it rather than reproduce it.

Exhaustive enumeration. `scratch/2026-10-22-mirror-LA-verify.py` (exact integer / permutation arithmetic). Over all monoids of order ≤ 3 and all their normalizers: (mLA) $\Rightarrow$ ($\mathsf{RA}_\#$), (mLA) $\Rightarrow$ associative, and (mLA) $\Leftrightarrow\!\sim_K$ left-congruence hold with zero exceptions; right-congruence never fails under (LA); and exactly 2 separators ($\mathsf{RA}_\#$) $\wedge \neg$(mLA) occur, both at order 3 (none at order ≤ 2). Over all (H, T) transversal pairs of S_3 (15) and S_4 (35 416): (LA) never fails, [(mLA) $\Leftrightarrow$ H normal] has zero mismatches, and [$(T, \#)$ associative $\Leftrightarrow$ the criterion] has zero mismatches. The explicit S_3 witness of the "closed but (mLA) fails" corner is $H = \langle(1\,2)\rangle$ (order 2, non-normal), $T = A_3$ (a subgroup, hence closed): (LA) ✓, (mLA) fails, $(T, \#) \cong C_3$ associative.

Honest scope. Theorems 2.1–4.2 are proved by hand (elementary: (K1), (LA), (mLA), associativity of $\cdot$, and, for §4, the right-coset structure); the enumerations corroborate but are not the proof. Minimality of the order-3 separator is a computational fact (exhaustive over order ≤ 3), as in the closure note, and nothing downstream depends on it. The group reading assumes a right transversal with $\mathrm{id} \in T$; the left-transversal reading swaps (LA) $\leftrightarrow$ (mLA) and gives the mirror statements. The group specialization (Theorem 4.1) and the S_3/S_4 enumerations are Python-only (permutation searches, not reducible kernel `decide`s at that scale); the Lean leaf certifies the abstract finite core.

What this settles

The two associative families are one mechanism and its two classical degenerations. Whenever the kernel of K is a two-sided congruence — i.e. (LA) $\wedge$ (mLA) — the magmoid is the honest quotient, $\mathbf{Norm}_K \cong \mathcal{A}/\!\sim_K$. (mLA) is a legible, checkable sufficient condition for associativity ("the kernel is a left congruence"), strictly stronger than bare ($\mathsf{RA}_\#$) and incomparable with closure of $\mathrm{Im}\,K$; the two sit in a diamond over associativity. Over groups, (mLA) is normality and closure is subgroup-hood, and both are solutions of the single conjugation criterion $f\varphi(g,h)f^{-1} \in H$ —

which also exhibits, between them, the genuine associative-without-either regime. This is exactly the unification SHOULD S1(b) asked for.

References

[1] MacBeth, *The positive home of the normalization defect: a one-sided-absorbing idempotent normalizer* (program note, 2026-10-19; registry node `positive-home-one-sided-absorbing-idempotent`). Source of Theorems A and B and the handedness definition recalled here.

[2] MacBeth, *What normalization's non-associativity is, and is not: three obstruction theories that miss it, and the idempotent normalizer that receives it* (triangulation unit, `expository/2026-10-21-normalization-defect-triangulation`, content commit `75a0756`). Contains the closure criterion and the provenance discussion (Baer; Lal; Kiechle; Grabowski).

[3] E. Di Lavore, M. Román, P. Székely, *Normalization is a sesquilaw*, arXiv:2510.01131. Origin of the magmoid **Norm** and the renormalization idempotent.

[4] R. Baer, Nets and groups, *Trans. Amer. Math. Soc.* **46** (1939) 110–141.

[5] R. Lal, Transversals in groups, *J. Algebra* **181** (1996) 70–81.

[6] H. Kiechle, *Theory of K-Loops*, Springer LNM 1778 (2002).

The oidified (categorical) version of this transversal/right-loop construction is published (Grabowski, *An introduction to loopoids*): see the provenance discussion of the triangulation unit [2], whose source index logs it. It is not re-cited here directly pending a deep-read registration.

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